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Electrode design for negative dielectrophoresis

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Abstract. A simple and practical procedure is described for the design of electrodes that produce well defined non-uniform electric fields for the study and application of negative dielectrophoresis. Such electrode systems can be used to direct and assemble colloidal particles away from electrode edges, and this is demonstrated for the case of yeast cells.

1. Introduction

As first defined by Pohl (1951, 1978) the term *dielectrophoresis* is used to describe the ponderomotive force exerted by a non-uniform electric field on polarized, but electrically uncharged, particles. The practical applications of dielectrophoresis include particle separation on the basis of differential polarizability (Benguigui *et al* 1986), the capture and retention of dielectric and conducting particles (Shalom and Lin 1985), particle levitation (Jones and Kallio 1979), the movement of cells using non-uniform travelling fields (Masuda *et al* 1988) and the alignment of cells prior to their electrofusion (Zimmermann and Vienken 1982, Washizu 1990). Measurement of the dielectrophoretic force has also been applied to the dielectric characterization of materials (Mognaschi and Savini 1983) and biological cells (Burt *et al* 1989, 1990, Chizmadzher *et al* 1989, Price *et al* 1988), and also to the behaviour of microbubbles in insulating liquids under high electrical stress (Feeley and McGovern 1988).

The general phenomenological equation describing the dielectrophoretic (DEP) force F acting on a particle of volume v in a non-uniform electric field is given by

$$F = \frac{1}{2}pv\nabla|E|^2 \quad (1)$$

where p is the nett effective polarizability of the suspended particle, E is the local root mean square (RMS) electric field and ∇ is the del vector operator. The dependence on E^2 indicates that the force direction is independent of the polarity of the applied field E and thus both DC and AC fields can be used. If the particle is surrounded by or immersed in a medium whose complex conductivity (or permittivity) is less than that of the particle, then the factor p in equation (1) is positive and the resulting DEP force will be directed *towards* the high-field region. This is known as *positive* dielectrophoresis. For the case where the complex conductivity of the particle is less than that of the surrounding medium, a

negative DEP force will result that is directed *away* from the high-field region.

The existence of isolated electric field maxima is not possible in a three-dimensional, curl and source free, electric field (Jones and Bliss 1977) and thus particles experiencing a positive DEP force can only be directed towards, or collected at, electrode edges. For this reason the literature in this field is predominantly concerned with the phenomenon of *positive* dielectrophoresis. However, isolated electric field minima can occur and this allows for the possibility of negative DEP forces being used to *direct and collect particles away from electrodes*, which offers distinct advantages for many envisaged biotechnological applications.

For controlled and predictable applications of dielectrophoresis, a knowledge of the non-uniform electric field distribution established by the electrode geometry is desirable. Unfortunately, apart from cylindrically or spherically symmetrical fields and the isomotive field geometry (Pohl and Pethig, 1977), no analytical expressions are readily available for such field distributions. We report here a procedure for designing electrodes with defined analytical expressions for the spatial variation of the factor $\nabla|E|^2$ in equation (1). Conclusions of practical importance are discussed and the experimental results obtained using one of the derived electrode geometries are described.

2. Theory

The electrode design procedure is based on the assumption that the electrical potential at any point (x,y,z) created by an electrode system of interest is defined by a polynomial that obeys Laplace's equation. By substituting this polynomial into Laplace's equation the corresponding equipotentials can therefore be determined, and these in turn can be used to define the required electrode boundaries. Although the method can readily

be applied to three-dimensional systems, for the sake of simplicity the treatment described here is restricted to two dimensions.

As an example of the method we shall consider an electric potential defined by a third-order polynomial of the form

$$f_3(x,y) = a_1x^3 + a_2x^2y + a_3xy^2 + a_4y^3 \quad (2)$$

where the subscript 3 in $f_3(x,y)$ indicates the order n . In general, there will be a total number C_{n+1}^n (here C_4^3) of coefficients (a_1 to a_4). The second-order partial derivatives are:

$$\begin{aligned} \partial^2 f_3(x,y)/\partial x^2 &= 6a_1x + 2a_2y \\ \partial^2 f_3(x,y)/\partial y^2 &= 2a_3x + 6a_4y. \end{aligned} \quad (3)$$

As $f_3(x,y)$ is defined as a solution of Laplace's equation, i.e.

$$\partial^2 f_3(x,y)/\partial x^2 + \partial^2 f_3(x,y)/\partial y^2 = 0$$

then

$$6a_1x + 2a_2y + 2a_3x + 6a_4y = 0 \quad (4)$$

is valid for any values of x, y . Thus, the coefficients a_i ($i=1, 2, 3, 4$) should satisfy the constraints:

$$a_3 = -3a_1 \quad a_2 = -3a_4 \quad (5)$$

and $f_3(x,y)$ can generally be expressed as

$$f_3(x,y) = a_1x^3 - 3a_1xy^2 + a_4y^3 - 3a_4x^2y \quad (6)$$

where a_1 and a_4 are independent parameters and $f_3(x,y)$ is the linear combination of the two independent functions:

$$f_{3a} = x^3 - 3xy^2 \quad f_{3b} = y^3 - 3x^2y.$$

Since f_{3a} and f_{3b} transpose into each other on interchanging the X - Y axes, then only one of them (e.g. f_{3a}) needs to be determined. Also, since the electric potential function based on f_{3a} is $f_3(x,y) = a_1(x^3 - 3xy^2)$, it can be used to define equipotentials in the electric field. For example, equipotentials can be constructed according to the equations

$$\begin{aligned} x^3 - 3xy^2 &= k \\ x^3 - 3xy^2 &= -k \end{aligned} \quad (7)$$

and these are shown (for positive and negative polarities) in figure 1.

If the electrodes are shaped according to the equipotential lines shown in figure 1, and positive and negative potentials of $+v$ and $-v$ are applied, then the potential $f_3(x,y)$ at any point (x,y) in the inter-electrode space is given by:

$$f_3(x,y) = \frac{v}{k}(x^3 - 3xy^2). \quad (8)$$

For arbitrary applied electrode potentials v_1 and v_2 , the corresponding potentials are given by

$$f_3(x,y) = \frac{v_2 - v_1}{2k}(x^3 - 3xy^2) + \frac{v_2 + v_1}{2}. \quad (9)$$

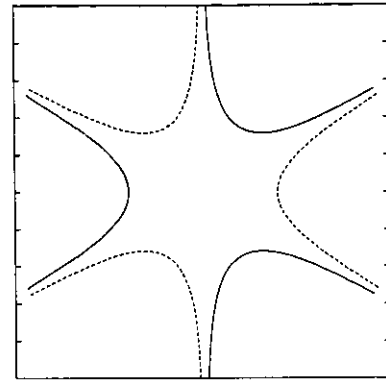


Figure 1. Equipotentials constructed from equations (7), where the full and broken curves refer arbitrarily to positive and negative polarities.

The constant k in equation (7) used for the determination of the equipotentials can be chosen arbitrarily and will not affect the final electric field pattern established by the electrode, but its value does determine the size of the designed electrode. (The closest approach of the electrode edges to the origin $(0,0)$ lies on a circle of radius $k^{1/n}$.)

Thus, in general, the polynomials defining the electric potential are of the form

$$f_n(x,y) = af_{na} + bf_{nb}$$

and these are given in table 1 for the cases $n=1, 2, 3$ and 4. Each $f_n(x,y)$ is the sum of two independent functions f_{na} and f_{nb} , whose form depends on the choice of the X - Y axes. As described above, these functions can be used to define the electrode boundaries, and examples of these are shown in figure 2 for $n=1, 2, 3$ and 4. The result obtained for $n=1$ gives the electrode boundaries as two parallel lines, and the resultant electric field is a uniform one. For $n>1$, the electrodes ($2n$ in number) produce non-uniform fields.

The spatial variations of the electric field $|E|$ and the factor $\nabla|E|^2$ of equation (1) produced by the various 'polynomial' electrodes are given in table 2. The electric field components E_x and E_y were determined by partial differentiation of the relevant potential functions $f_n(x,y)$ having the form as shown for $n=3$ in equation (9). For $n=1$, the electric field is uniform, $\nabla|E|^2=0$ and no DEP force is exerted on any particles located between the electrodes. For $n>1$ then $\nabla|E|^2$ is a function of position (x,y) and n , and in general varies as $(x^2 + y^2)^{(n-1.5)}$. Three-dimensional plots of the electric field strength $|E|$ and the magnitude of $\nabla|E|^2$ as a function of position

Table 1. Polynomials derived from Laplace's equations for defining electrode geometry.

n	$f_n(x,y) = af_{na} + bf_{nb}$
1	$ax + by$
2	$a(x^2 - y^2) + bxy$
3	$a(x^3 - 3xy^2) + b(y^3 - 3x^2y)$
4	$a(x^4 - 6x^2y^2 + y^4) + b(x^3y - xy^3)$

Table 2. Spatial variation of the field $|E|$ and DEP force factor $\nabla|E|^2$ for the polynomial electrodes.

n	$ E = (E_x^2 + E_y^2)^{0.5}$	$\nabla E ^2$
1	$\frac{V_2 - V_1}{2k_1}$	0
2	$2 \frac{V_2 - V_1}{2k_2} (x^2 + y^2)^{0.5}$	$8 \left(\frac{V_2 - V_1}{2k_2} \right)^2 (x^2 + y^2)^{0.5}$
3	$3 \frac{V_2 - V_1}{2k_3} (x^2 + y^2)$	$36 \left(\frac{V_2 - V_1}{2k_3} \right)^2 (x^2 + y^2)^{1.5}$
4	$4 \frac{V_2 - V_1}{2k_4} (x^2 + y^2)^{1.5}$	$96 \left(\frac{V_2 - V_1}{2k_4} \right)^2 (x^2 + y^2)^{2.5}$
n	$n \frac{V_2 - V_1}{2k_n} (x^2 + y^2)^{0.5(n-1)}$	$2(n-1)n^2 \left(\frac{V_2 - V_1}{2k_n} \right)^2 (x^2 + y^2)^{(n-1.5)}$

V_1, V_2 : applied electrode potentials.
 k_1, k_2, k_3 and k_4 : constants defining the electrode size. See text for details.

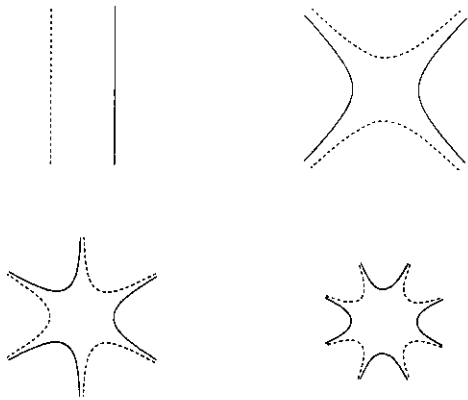


Figure 2. Boundaries of the 'polynomial' electrodes for $n=1, 2, 3$ and 4 , where the full and broken curves denote differences in electrode polarity. The number of electrodes of each polarity is equal to n .

(x,y) are given in figure 3 for a third-order polynomial electrode. These plots clearly show the location of an isolated field minimum, where in fact the field is zero at the centre point $(0,0)$ and where the DEP force is zero.

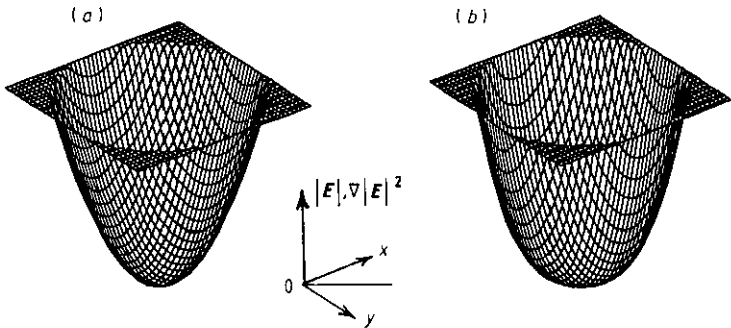


Figure 3. Three-dimensional plots of (a) the electric field $|E|$ and (b) the dielectrophoretic force factor $\nabla|E|^2$ for the third-order ($n=3$) polynomial electrode.

Particles of a lower effective polarizability than that of the surrounding suspending medium will experience a negative dielectrophoretic force and be directed towards the point $(0,0)$.

It is of value to determine how the spatial variation of the DEP force varies according to the order of the polynomial electrode design. This can conveniently be accomplished by comparing the factor $\nabla|E|^2$ for various values of n as a function of x and y for the same value of the inner radius that defines the closest approach of the electrode edges to the centre point $(0,0)$. This comparison is shown in figure 4 for an inner radius of $20\text{ }\mu\text{m}$ and for a potential difference of 20 V applied to the corresponding electrode pairs. The factor $\nabla|E|^2$ varies as r^{2n-3} , where r is the radial distance from the centre $(0,0)$ to the point of interest ($r=(x^2 + y^2)^{0.5}$). Plots of the form of figure 4 are of value in optimizing negative dielectrophoretic collection, or particle separation through differential polarizability, according to particle size and concentration.

3. Experiment

Electrodes of 'polynomial' geometry, corresponding to $n=2$ and $k=4096\text{ }\mu\text{m}^2$, were produced using standard

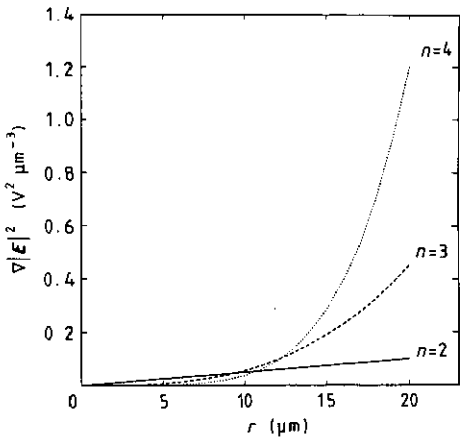


Figure 4. Variation of the factor $\nabla|E|^2$ as a function of distance r from the centre of the electrode geometry for different values of the polynomial order n .

photolithographic procedures and are shown magnified in figure 5. The electrode material was gold, 70 nm thick, vacuum evaporated onto a 5 nm thick seed layer of chromium on glass. For $n=2$, a value of the geometry factor k of $4096 \mu\text{m}^2$ produces electrodes whose tips lie on a circle of radius $64 \mu\text{m}$ about the centre point, and which approach a parallel spacing of $38 \mu\text{m}$ at distances far from the centre.

Yeast cells (*Saccharomyces cerevisiae*, strain RXII) were inoculated in plain broth and shaken for 48 h in a water bath at 303 K and then stored for seven days at 277 K. Cells were harvested from the growth medium by centrifugation (100 g for 5 min), washed three times in 280 mM mannitol and finally suspended in 280 mM mannitol. The concentration of suspended cells was determined from optical absorption measurements, as detailed elsewhere (Burt *et al* 1989). Figure 5 shows yeast cells, at a concentration of 2.7×10^7 cells per ml, suspended in 280 mM mannitol solution within the volume defined by the gold 'polynomial' electrodes. The conductivity of the yeast suspension was measured to be $3.61 \mu\text{S cm}^{-1}$.

Figure 6 shows positive dielectrophoretic collection of the yeast cells at the electrode edges, after application of a 10 kHz, 10 V peak-peak voltage to the electrodes.

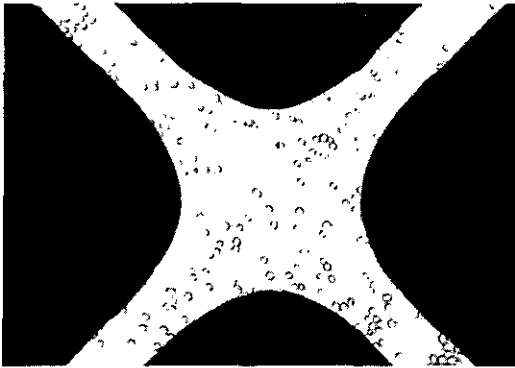


Figure 5. Yeast cells suspended between electrodes of polynomial geometry corresponding to $n=2$ and $k=4096 \mu\text{m}^2$.

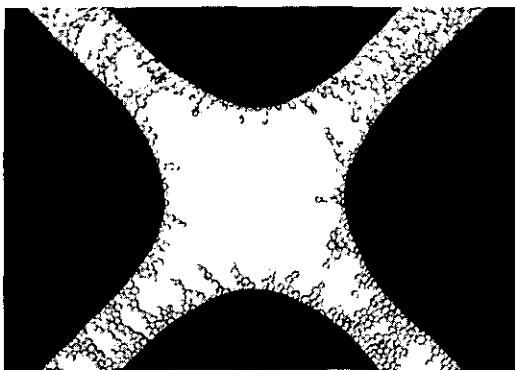


Figure 6. Positive dielectrophoretic collection of yeast cells suspended in 280 mM mannitol of conductivity $3.61 \mu\text{S cm}^{-1}$, with 10 kHz, 10 V peak-peak signal applied to electrodes.

Figure 7 shows the result of applying a similar 10 kHz signal to the electrodes, for the case where the yeast cells had been suspended in 280 mM mannitol solution to which had been added sufficient KCl to raise the suspension's conductivity to $170 \mu\text{S cm}^{-1}$. The yeast cells are seen to have been directed to the central region between the electrodes. In other words, they had experienced *negative* dielectrophoresis. On applying a rotating electric field to the electrodes, using a field generator designed and constructed by Hölzel in our laboratory (Hölzel 1988), the negative dielectrophoresis mode was found to offer the advantage of stabilizing the position of particles when exhibiting contrafield electro-rotation (Arnold and Zimmermann 1988).

4. Conclusions

The procedure described here can be used to design microelectrodes that provide well defined spatial variations of non-uniform electric fields. Such electrodes can be used in applications that exploit the phenomena of dielectrophoresis, and in particular for directing and concentrating particles away from electrode edges using *negative* dielectrophoretic forces. The ability to control the effect of negative dielectrophoresis can also be used to stabilize the location of particles that are subject to contrafield electro-rotation forces.

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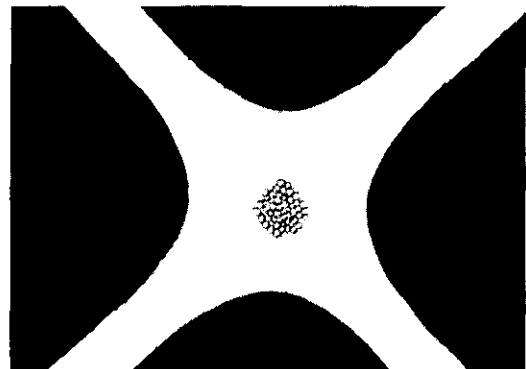


Figure 7. Negative dielectrophoretic collection of the same yeast cells as shown in figures 5 and 6, when suspended in a 280 mM mannitol + 1.4 mM KCl solution of conductivity $170 \mu\text{S cm}^{-1}$ and subjected to a 10 kHz, 10 V peak-peak signal applied to the electrodes.

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